Setup Guide

1. Install python 3.13

Visit the official Python download page:

👉 https://www.python.org/downloads/

Download the latest **Python 3.x Windows Installer** (usually ends with .exe).

Run the installer:

Check **“Add Python to PATH”** (very important).

Click on **“Customize installation”** if you want to change install location or install for all users.

Click **“Install Now”**.

Verify installation:

Open **Command Prompt** and run:

python --version

pip --version

where python → will give path of your python exe file.

2. Setting Up the Python PATH

If you did **not** check "Add Python to PATH" during installation, you must add it manually:

Find the Python Installation Directory:

Default location:

C:\Users\<your username>\AppData\Local\Programs\Python\Python3x

or run command ‘where python’ on command prompt. It will give python.exe path

Replace < your username> with your Windows username, and python3 with your installed version (e.g.,python 312)

Add Python to PATH:

Open the **Start** menu, search for "Environment Variables," and select **Edit the system environment variables**.

In the System Properties window, click **Environment Variables**.

Under **User variables** or **System variables**, select Path ****and click Edit.

Click **New** and paste the path to your Python directory (where python.exe is located).

Click **OK** to close all dialogs.

Verify Installation

Open Command Prompt and run:

python --version

3. Install Google Chrome (If Not Already Installed)

Download Chrome:

Official download page:

https://www.google.com/chrome/[4](https://support.google.com/chrome/answer/95346?co=GENIE.Platform%3DDesktop)

Follow the on-screen instructions to download and install Chrome.

4. Install ChromeDriver

Check Chrome Version:

Open Chrome, go to the menu (three dots) > **Help** > **About Google Chrome**.

Note the version number (e.g., 136.0.7103.93).

Download ChromeDriver

Go to the official ChromeDriver site:

https://sites.google.com/chromium.org/driver/

Download the ChromeDriver version that matches your Chrome browser version.

Choose the correct platform (e.g., "chromedriver-win64.zip").

Install ChromeDriver

Extract the chromedriver.exe file from the ZIP archive.

For above mentioned chrome version, chromedriver can be downloaded from

For win64: curl https://storage.googleapis.com/chrome-for-testing-public/136.0.7051.0/win64/chromedriver-win64.zip

For win64: curl https://storage.googleapis.com/chrome-for-testing-public/136.0.7103.94/win64/chromedriver-win64.zip

For wind32: curl https://storage.googleapis.com/chrome-for-testing-public/136.0.7103.94/win32/chromedriver-win32.zip

Place the chromedriver.exe in a known folder (e.g., c:\tools\chromedriver)

Add ChromeDriver to PATH:

Follow the same steps as for Python, but add the folder containing chromedriver.exe to your PATH variable.

5. Python project setup

Create a folder where you want to keep your files.

Create virtual environment inside the folder.

Commands you can use:

python3 -m venv <env name>

source <env name>/bin/activate

pip install -r requirements.txt → basic requirements.txt is available

To use created environment with your Jupyter notebook, use below commands: pip install ipykernel python3 -m ipykernel install --user --name <myname>

Define project structure and here you start coding 🙂

6. Milvus Setup (can be done later)

Install docker using below command

snap install docker

Get docker compose

wget https://github.com/milvus-io/milvus/releases/download/v2.4.0/milvus-standalone-docker-compose.yml -O docker-compose.yml

Make Sure Docker and Docker Compose Are Installed

Check Docker:

docker --version

Docker version 28.1.1, build 068a01e

Check Docker Compose

docker compose version

Docker Compose version v2.33.1

(If you get a version, you're good. If not, install Docker Compose.)

Before starting milvus to give customized path for data volumes:

export DOCKER_VOLUME_DIRECTORY=/home/<user>/Projects/pdf_to_markdown/milvus

Start Milvus with Docker Compose

In the directory where your docker-compose.yml is saved, run:

docker compose up -d

The d flag runs Milvus in detached (background) mode.

Check That Milvus Is Running

See running containers:

docker ps

You should see containers for Milvus, etcd, and MinIO.

(Optional) Check Milvus health

curl localhost:9091/api/v1/health

You should get a response like: {"status": "ok"}

Connect to Milvus from Python

Now you can use pymilvus to connect:

```
from pymilvus import connections
    connections.connect(host="localhost", port="19530")
```

Stop Milvus When Done

To stop the Milvus server:

docker compose down